

ATMORE AGDM, AB, Canada

WMO: 712870

Lat: 54.7802N

Lon: 112.8248W

Elev: 574

StdP: 94.62

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-33.5	-29.5	-37.0	0.1	-33.3	-32.8	0.2	-29.1	10.0	1.9	8.5	-1.4	1.2	230	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	27.8	18.0	26.0	16.8	24.3	16.2	19.3	25.9	18.1	24.1	17.1	22.6	3.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.9	22.3	15.7	12.0	20.9	14.7	11.2	19.7	57.5	26.0	53.4	24.1	50.1	22.6	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.4	6.4	-39.0	31.0	3.4	1.9	-41.4	32.4	-43.4	33.4	-45.4	34.5	-47.8	35.9		
(5)			-38.9	21.5	3.4	1.3	-41.3	22.4	-43.3	23.1	-45.2	23.8	-47.7	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-12.2	-11.7	-4.9	3.2	9.9	13.9	16.4	14.7	9.8	3.2	-5.9	-12.7	(6)
(7)		DBStd	12.12	8.68	8.05	7.27	5.17	4.36	2.94	2.72	3.25	4.26	4.65	6.76	7.97	(7)
(8)		HDD10.0	3477	687	607	462	208	56	4	0	3	54	214	477	705	(8)
(9)		HDD18.3	5961	945	840	720	453	263	134	73	119	255	468	727	963	(9)
(10)		CDD10.0	580	0	0	0	5	52	122	199	148	49	4	0	0	(10)
(11)		CDD18.3	22	0	0	0	0	1	3	13	6	1	0	0	0	(11)
(12)		CDH23.3	579	0	0	0	1	44	90	253	144	43	4	0	0	(12)
(13)		CDH26.7	103	0	0	0	0	7	11	56	22	7	0	0	0	(13)
(14)	Wind	WSAvg	2.7	2.6	2.5	2.9	3.3	3.2	2.9	2.6	2.5	2.7	2.8	2.6	2.4	(14)
(15)	Precipitation	PrecAvg	405	17	12	16	24	41	70	78	52	33	18	16	17	(15)
(16)		PrecMax	507	39	27	34	57	86	138	155	90	74	56	41	39	(16)
(17)		PrecMin	300	0	0	3	3	6	22	22	21	9	5	5	0	(17)
(18)		PrecStd	61	11	7	9	13	19	30	27	19	18	11	9	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	7.3	11.9	22.4	27.4	28.1	30.8	29.1	27.8	22.6	10.9	6.3	(19)
(20)			MCWB	3.0	2.8	5.8	10.8	14.3	17.4	19.9	18.5	16.9	12.3	6.6	2.6	(20)
(21)		2%	DB	4.3	4.7	8.8	18.2	24.1	25.9	28.2	26.8	24.1	16.6	6.9	3.4	(21)
(22)			MCWB	1.7	1.4	3.9	8.8	12.7	16.1	19.0	18.1	15.3	9.8	3.4	0.8	(22)
(23)		5%	DB	2.3	2.2	6.7	15.2	21.7	23.9	26.0	24.8	21.3	13.4	4.5	0.6	(23)
(24)			MCWB	0.4	-0.3	2.6	7.1	11.7	15.3	17.7	17.0	13.8	7.5	2.0	-1.1	(24)
(25)		10%	DB	0.0	0.1	4.7	12.5	19.7	21.8	24.2	22.9	18.3	10.9	2.2	-1.9	(25)
(26)			MCWB	-1.4	-1.6	1.2	5.7	10.5	14.3	17.0	16.2	12.3	6.2	0.4	-3.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	3.2	6.1	11.2	15.1	18.8	21.4	20.6	17.5	12.8	6.6	2.9	(27)
(28)			MCD B	6.0	6.5	11.2	21.4	24.3	26.1	27.9	26.5	25.8	21.4	10.8	5.7	(28)
(29)		2%	WB	2.0	1.6	4.2	9.1	13.7	17.3	19.9	18.9	15.8	10.1	4.0	0.8	(29)
(30)			MCD B	4.2	4.4	8.3	17.4	22.3	23.4	26.9	25.2	22.7	15.4	6.1	3.0	(30)
(31)		5%	WB	0.5	0.1	2.8	7.5	12.4	16.2	18.6	17.6	14.3	8.1	2.2	-1.1	(31)
(32)			MCD B	2.3	2.1	6.4	14.3	20.1	22.1	24.7	23.4	20.2	12.6	4.1	0.4	(32)
(33)		10%	WB	-1.5	-1.7	1.5	6.0	11.2	15.1	17.6	16.6	12.8	6.7	0.4	-3.0	(33)
(34)			MCD B	0.0	0.2	4.3	11.8	18.0	20.5	22.9	22.1	17.4	10.3	2.1	-1.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	12.3	11.7	12.2	14.1	12.9	13.4	13.6	13.2	10.9	8.9	9.7	(35)
(36)			MCD BR	10.7	14.2	12.8	17.3	18.1	16.8	16.8	16.9	18.4	16.1	11.2	11.3	(36)
(37)			MCWBR	8.4	11.3	8.5	9.5	8.8	8.3	8.6	8.8	10.0	9.6	7.9	9.3	(37)
(38)		5% WB	MCD BR	10.5	13.7	12.2	16.2	16.3	14.5	15.3	15.2	17.2	14.9	10.4	11.0	(38)
(39)			MCWBR	8.4	11.1	8.1	9.3	8.2	7.9	8.3	8.4	9.6	9.2	7.7	9.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.250	0.275	0.301	0.330	0.342	0.355	0.357	0.365	0.313	0.283	0.252	0.234	(40)	
(41)		taud	2.307	2.307	2.348	2.390	2.413	2.381	2.435	2.355	2.518	2.571	2.465	2.301	(41)	
(42)		Ebn at Noon	716	821	880	898	903	890	882	850	863	820	729	658	(42)	
(43)		Edh at Noon	53	76	93	103	107	112	104	106	79	59	46	42	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.80	1.80	3.29	4.62	5.63	5.85	5.89	4.84	3.34	1.85	0.85	0.57	(44)	
(45)		RadStd	0.09	0.13	0.27	0.37	0.50	0.40	0.32	0.35	0.34	0.26	0.09	0.04	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page